MGU-UGP (HONOURS) REGULAR EXAMINATION – March 2026
Fourth Semester
Discipline Specific Core Course (DSC) - MG4DSCCSC201 - COMPUTER NETWORKS
(2024 Admission Onwards)

Scheme
Duration: 2 Hours **Maximum Marks: 70**

Part A – Very Short Answer Questions
Answer all questions
Each question carries 2 marks

1. Define syntax and semantics in a network protocol.
Ans. Syntax. The term syntax refers to the structure or format of the data, meaning the order in which they are presented.
Semantics. The word semantics refers to the meaning of each section of bits.

2. List any two characteristics of circuit switching.
Ans. In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase. Data transferred between the two stations are not packetized (Any relevant two points)

3. Identify one technique used for error correction.
Ans. Hamming Code (Forward Error Correction technique).

4. State the range of port numbers available in UDP.
Ans. It is 16 bits long, which means that the port number can range from 0 to 65,535.
Range of port numbers in UDP
- 0 to 65535
  - Well-known: 0–1023
  - Registered: 1024–49151
  - Dynamic: 49152–65535

5. Define LCP in PPP.
Ans. The Link Control Protocol (LCP) is responsible for establishing, maintaining, configuring, and terminating links. It also provides negotiation mechanisms to set options between the two endpoints. Both endpoints of the link must reach an agreement about the options before the link can be established. [2 × 5 = 10]

Part B – Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

6. Differentiate between LAN and WAN.
Ans. Local Area Network (LAN) and a Wide Area Network (WAN) differ mainly in terms of geographical coverage, ownership, purpose, and structure.
A LAN is a privately owned network that connects devices within a limited geographical area such as a single office, building, or campus. It is primarily designed to share resources like

hardware (printers), software (applications), and data among computers or workstations. A LAN can be as simple as a few devices or as large as an entire organization, but its size is typically limited to a few kilometers. LANs usually use a single transmission medium and follow common topologies such as bus, star, or ring. They provide high data rates, typically ranging from 100 Mbps to 1000 Mbps, ensuring fast communication.

In contrast, a Wide Area Network (WAN) covers a very large geographical area, which may extend across a country, continent, or even the entire world. Unlike LANs, WANs are not usually privately owned; they often rely on public or leased communication infrastructure such as telephone lines or satellite links. The main purpose of a WAN is to connect multiple LANs and remote systems. WANs can be of two types: switched WANs, which use routers to connect networks (e.g., the Internet backbone), and point-to-point WANs, which use leased lines to connect a user or small network to an Internet Service Provider (ISP). WANs are generally more complex, have lower data rates compared to LANs, and involve technologies such as Frame Relay, ATM, and wireless WANs.

In summary, while LANs are small, fast, and privately managed networks focused on resource sharing, WANs are large-scale networks designed for long-distance communication and interconnection of multiple networks.

(Any 5 relevant differences – 1 mark for each difference)

7. Explain the working of a connection-oriented service.

Ans. Concept – 1 mark

Example of how Circuit switching works – 3 marks

Features of Connection oriented service – 1 mark

8. Describe the features of IPv4 protocol.

Ans. Internet Protocol version 4 (IPv4) is the fundamental protocol used for delivering packets in a TCP/IP network. It provides a mechanism for logical addressing and routing of data across interconnected networks.

One of the key features of IPv4 is that it is a connectionless protocol. This means that it does not establish a dedicated path before transmitting data. Each packet, known as a datagram, is sent independently and may follow different routes to reach the destination.

IPv4 is also an unreliable protocol because it does not guarantee the delivery of packets. Data may be lost, duplicated, or arrive out of order. It follows a best-effort delivery approach, where it attempts to deliver packets but provides no assurance of success. This is similar to a postal service that tries to deliver mail but cannot guarantee delivery.

Another important feature is that IPv4 provides no flow control or error control, except for basic error detection in the header. It does not handle congestion or retransmission of lost packets. Instead, it relies on higher-level protocols such as TCP to ensure reliability and proper sequencing.

IPv4 uses a datagram-based packet switching technique, where each packet is treated independently. This allows efficient routing but may lead to packets arriving in different orders or experiencing varying delays.

Finally, IPv4 provides logical addressing using 32-bit addresses, which uniquely identify devices on a network and enable routing across multiple interconnected networks.

In summary, IPv4 is a connectionless, unreliable, best-effort protocol that enables packet delivery using logical addressing, while relying on higher-layer protocols for reliability and control.

9. Explain the working of TELNET in a computer network.

Ans.

Definition and Purpose - 2 marks

TELNET (Telecommunication Network) is an application layer protocol in the TCP/IP suite that provides a remote login service. It allows a user to log into another computer across the network and work as if locally connected.

Working – 3 marks

10. Outline the layered architecture of the OSI model.

Ans. Layered Architecture of the OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework that standardizes the functions of a communication system into seven distinct layers. Each layer performs specific tasks and provides services to the layer above it while receiving services from the layer below. This layered approach simplifies network design, implementation, and troubleshooting. The seven layers of the OSI model are:

1. Physical Layer – Responsible for the transmission of raw bits over a physical medium. It deals with electrical, mechanical, and procedural specifications such as cables, connectors, and signaling.
2. Data Link Layer – Ensures reliable data transfer by organizing bits into frames. It handles error detection, flow control, and physical addressing.
3. Network Layer – Manages logical addressing and routing of packets from source to destination across multiple networks.
4. Transport Layer – Provides end-to-end communication, ensuring complete data transfer with error control, flow control, and segmentation.
5. Session Layer – Establishes, maintains, and terminates communication sessions between applications.
6. Presentation Layer – Translates data into a format suitable for the application layer. It handles encryption, compression, and data formatting.
7. Application Layer – Provides network services directly to end users, such as email, file transfer, and web access.

The OSI model follows a layered architecture, where communication between corresponding layers in different systems occurs logically (peer-to-peer), while actual data transmission flows through all layers. This structure promotes interoperability, modularity, and standardization in network communication.

11. Explain how a datagram network delivers packets without a dedicated path.

Ans. In a datagram network, packets are delivered without establishing a dedicated path between the source and destination. This type of network follows a connectionless approach, meaning there are no setup or teardown phases before or after transmission.

When a message is sent, it is first divided into smaller packets called datagrams. Each datagram is treated as an independent unit, even if it is part of the same message. Every packet carries the complete destination address, allowing it to be routed individually through the network.
In packet-switched networks, there is no resource reservation. This means that:

- No bandwidth is reserved in advance
- No fixed processing time is allocated
- Resources are assigned on a first-come, first-served basis

As each packet travels through the network, routers (packet switches) make independent forwarding decisions based on current network conditions. Because of this, packets from the same message may take different paths to reach the destination.
Due to the absence of a dedicated path and resource reservation:

- Packets may arrive out of order
- Packets may experience different delays
- Some packets may be lost or dropped due to congestion

The network itself does not ensure reliability. Instead, upper-layer protocols are responsible for:

- Reordering packets
- Detecting lost packets
- Requesting retransmission

In summary, a datagram network delivers packets independently without a predefined path, using dynamic routing and on-demand resource allocation, making it flexible but less reliable.

12. Show how Go-Back-N ARQ handles frame errors.

Ans. Go-Back-N ARQ is an error control technique used in data link layer protocols that allows the sender to transmit multiple frames before receiving acknowledgments, using a sliding window mechanism.
In this method, frames are sent sequentially and are numbered. The receiver only accepts frames in order and sends acknowledgments (ACKs) for correctly received frames.
When a frame error occurs (due to loss or corruption), the following process takes place:

1. The sender transmits multiple frames continuously within its window size.
2. If a frame is lost or damaged, the receiver detects the error and discards that frame and all subsequent frames, even if they are received correctly.
3. The receiver then sends an acknowledgment for the last correctly received frame (or may send a negative acknowledgment).
4. When the sender realizes that a frame has not been acknowledged (either through timeout or duplicate ACKs), it goes back and retransmits the erroneous frame and all frames sent after it.

This is why the method is called "Go-Back-N", as the sender goes back to the point of error and retransmits all subsequent frames. (3 marks)
Provide an example – 2 marks

13. Explain the role of protocol in a URL.

Ans. A Uniform Resource Locator (URL) is used to specify the address of a resource on the Internet. It consists of four main components: protocol, host, port, and path. Among these, the protocol plays a crucial role in determining how the resource is accessed.

The protocol in a URL defines the client-server communication method used to retrieve a document. It specifies the set of rules and procedures that govern how data is transmitted between the client (such as a web browser) and the server. In other words, the protocol tells the system how to communicate with the host computer to obtain the required information.

Different protocols are used depending on the type of service required. For example, HTTP (HyperText Transfer Protocol) is the most commonly used protocol for accessing web pages, while FTP (File Transfer Protocol) is used for transferring files. The protocol is always placed at the beginning of the URL, followed by a colon and slashes (e.g., http://), indicating the method to be used for data retrieval.

The choice of protocol affects how the data is handled, including aspects such as data format, connection type, and security. For instance, HTTP is used for standard web communication, while secure communication can use HTTPS.

In summary, the protocol in a URL specifies the type of service and communication rules required to access a resource, ensuring that the client and server interact correctly to retrieve the requested information. [5 × 6 = 30]

Part C – Essay Questions

Answer any 2 questions

Each question carries 15 marks

14. State the different types of network topologies with its relevant features.

Any 5 topologies with diagrams, features, advantages, limitations – 3 Mark each

Mesh, Star, Bus, ring, Hybrid, Tree etc

15. Explain the classification of routing algorithms.

Ans. Introduction to Routing concept, Routing algorithms – 4 marks

Classification –

Static vs Dynamic Routing – 2 marks

Intradomain Routing Protocols – 6 marks

Routing within a single autonomous system (AS).

Two major classes: Distance Vector Protocol (Routing Information Protocol (RIP).),

Link State Protocol (Open Shortest Path First (OSPF)

Interdomain Routing Protocols - 3 marks

Path Vector Protocol (Border Gateway Protocol (BGP))

16. Explain SMTP in detail.

Ans.

SMTP introduction – Role in email system – 5 marks

SMTP Architecture (UA, MTA, MAA)– 5 marks

Characteristics/features , limitations of SMTP – 5 marks [15 × 2 = 30]

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